

MUTUAL FUND ANALYTICS PLATFORM

Capstone Project I - Final Implementation Report

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GitHub Repository	https://github.com/Manimaranpro/bluestock_mf_capstone

1. Executive Summary

This report documents the end-to-end design and implementation of the Mutual Fund Analytics Platform. The objective of the platform is to ingest, clean, store, and analyze mutual fund transactional datasets, daily NAV history, and portfolio holdings. By utilizing advanced Python algorithms, relational database architectures, and analytical reports, the platform provides robust quantitative insights into portfolio performance and risk parameters.

2. Technical Architecture & ETL Pipeline

- **Ingestion & Cleaning:** Automated pipelines parsed and standardized date representations, applied forward-fill logic across holidays to eliminate missing valuations, and enforced integrity checks on financial variables.
- **Relational DB Migration:** Built a relational SQLite database (bluestock_mf.db) incorporating a 3NF Star Schema with strict primary/foreign key connections linking central fact tables to dimension tables.
- **SQL Analytics:** Drafted and verified 10 core analytical queries evaluating crucial business questions (including category averages, asset-under-management rankings, and expense limits).

3. Quantitative Portfolio Analytics

Using vectorized matrix calculations, the platform generated advanced investment risk-and-return metrics across all 40 schemes, avoiding slow nested iterations. The metrics include annualized Compound Annual Growth Rate (CAGR), Sharpe Ratios, Sortino Ratios, Maximum Drawdowns, and systematic risk coefficients (Alpha and Beta) calculated via OLS regressions against Nifty benchmarks.

4. Visual Findings & Analytical Artifacts

 NAV Trends Chart

Figure 1: Daily NAV Historical Trend (2022-2026)

Displays the daily NAV trajectory for schemes, highlighting market bull runs and corrections.

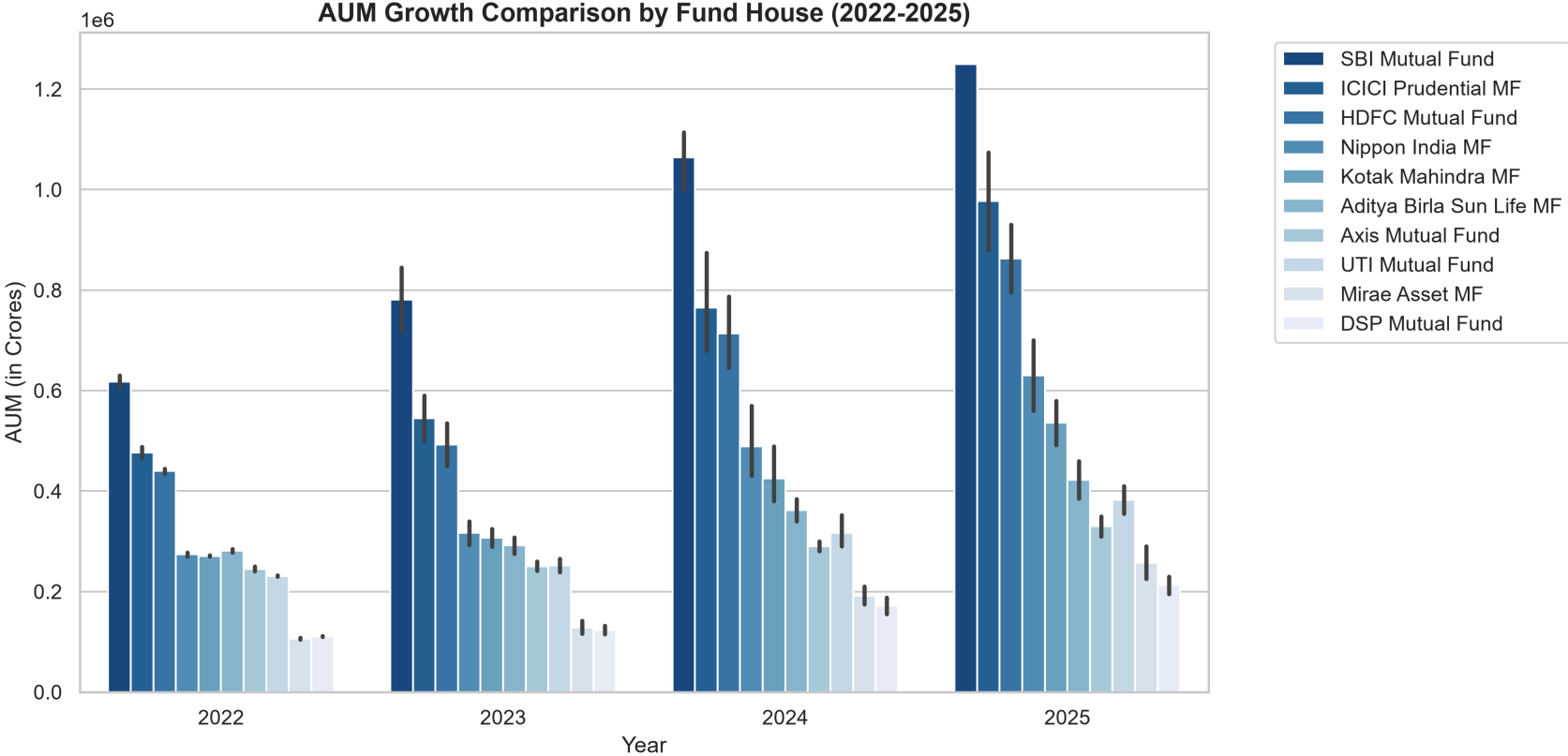


Figure 2: Annual AUM Growth Comparison

Compares Assets Under Management (AUM) across fund houses, illustrating market dominance.

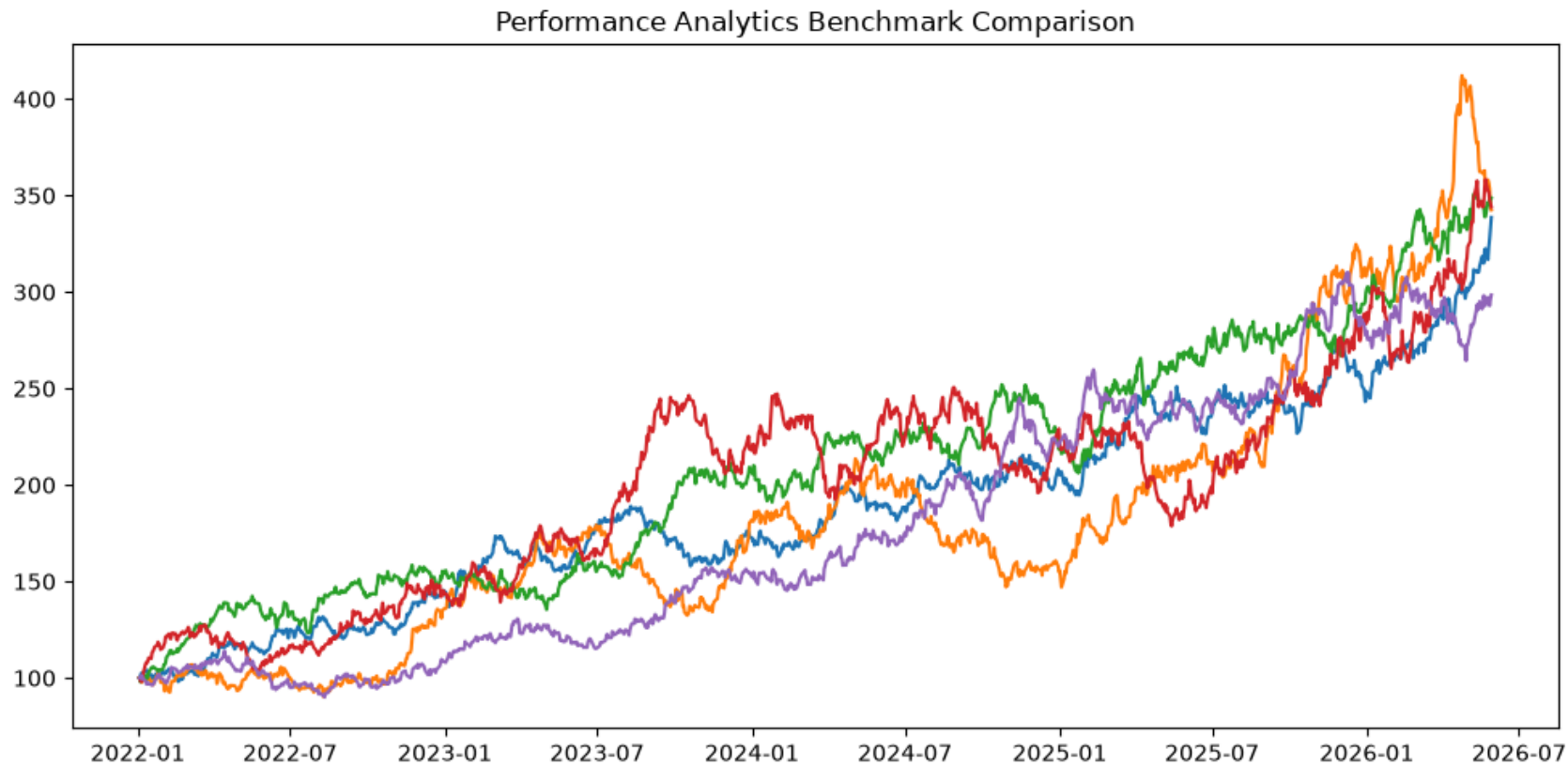


Figure 3: Top 5 Scorecard Funds vs. Nifty Benchmark

Tracks the normalized growth of the top 5 scorecard-selected funds against the market index.

5. Conclusion

The platform successfully automates the translation of raw financial transaction logs into structured, actionable business intelligence. The implementation of robust data structures, documented data dictionaries, and automated visual reports ensures the pipeline is reproducible, secure, and ready for production deployment.

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